

Chapter 4

The relationship between *u*-adjectives and *nu*-causatives in Anatolian

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The significant number of Hittite causative/factitive formations in *-nu-* attested next to *u*-adjectives has always been striking and led together with comparative evidence from Old Indic and Ancient Greek to a theory in the 70s that the process occurs through *n*-infixation between the root and the adjectival suffix *-u-*, e.g., Hitt. *tēp-u-* ‘small’ → Hitt. *tēp-n-u^{mi}* ‘to make small, diminish’ (matching OInd. **dabh-u-* (< PIE **d^heb^h-u-*, replaced by OInd. *dabh-rá-* ‘small’) → OInd. *dabh-n-ó-ti* ‘to harm, deceive’), Hitt. *park-u-* ‘high’ (< PIE **b^herǵ^h-u-*, beside Hitt. *park-taru* ‘must become high’) → Hitt. *parg-n-u^{mi}* ‘to make high’, provoking at a certain point a reanalysis of the morpheme *-n-* to *-nu-*. Although this theory never gained much popularity, recent studies on the Luwian language have brought forth more data that have yet to be examined in depth vis-à-vis the morphological process involved, e.g., HLuw. *wa/i-su-* /wasu-/ ‘good’ (< PIE **h₁ues-u-*, beside HLuw. *wa/i-s^o* /wass^{ti}/ ‘to be good, pleasing’) → HLuw. *u-sa-nú-* /usnuwaⁱ/ ‘to make good, benefit’, and, thus, a re-examination of the Anatolian material is warranted. In addition to retracing the development of the denominal morpheme *-nu-*, its extension to other types of nominal stems, notably to the *i*-stems (e.g., Hitt. *ḫarki-* ‘white’ (< PIE **h₂erǵ-i-*) → Hitt. *ḫarg(a)nu-* ‘to make white’), as well as the motivation behind it is addressed. Furthermore, the phenomenon of *u*-extended roots that has been considered to be intertwined with the development of the *nu*-suffix is explored and the discussion is expanded with a new set of Anatolian data (cf. HLuw. *wa/i₅-su-* /wasu-/ ‘to be good, successful’).

1 Introduction

The numerous examples of Hittite denominal factitives in *-nu-* had already been noticed in the 30s, when comparative linguistic work on Hittite was thriving.



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Indo-Europeanists were convinced that the suffix *-nu-* was primarily deverbal and that the denominal function had been secondarily acquired by Anatolian. In his doctoral thesis submitted in 1973, Harold Koch scrutinized the data more than it had been done before and cogently defended the view that Hittite factitives in *-nu-* were originally derived from *u*-adjectives via *n*-infixation. Further, he explained that the co-existence of root verbs, *u*-adjectives and *nu*-factitives all based on the same root would have been the starting point for the replacement of the isolated infixation process with the more common suffixation one.¹ Thus, by reanalysing the derivational base, the deverbal causative suffix would have been created.² This theory can be illustrated with the following examples, i.e., first the proto-type and second the reanalysis:

1. CVC-*u-* → CVC-*n-u-* (e.g., Hitt. *huišu-* ‘alive, raw’ → *huiš-n-u^{mi}* ‘to keep alive’)
2. CVC- → (reanalysis) CVC-*nu-* (e.g., Hitt. *huiš^{mi}* ‘to live’ → *huiš-nu^{mi}* ‘to keep alive, let live’)

It is noticeable that the formation of Hittite *huišnu^{mi}* is synchronically ambiguous, as one could even switch the example of the proto-type with the one of the reanalysis, arguing for the reverse direction in the development and making the *n*-infixation younger than the *nu*-suffixation.³ Koch (1973: 34–36), however, insists that the *n*-infixation is older by pointing out that an *n*-infixated verb like Hitt. *harni(n)k^{mi}* ‘to destroy’ is later replaced by the more productive suffixed formation Hitt. *harknu^{mi}* ‘to destroy’.⁴ The recently discovered Luwian verb

¹Koch 1973: 25–26, 34–36. Oettinger (1979: 165–166 with fn. 72) points out that the infixation process should be older than the apparent separation between nominal and verbal roots within Proto-Indo-European, adducing PIE **ued-* ‘water’ → **u-né-d-/und-* ‘to water’ (continued in Ved. *unátti*) as a case in point; cf. NIL 712, fn. 43. However, the nature of Proto-Indo-European roots and the relationship between diverse root formations is a topic that will require more attention in future research.

²This approach slightly differs from the theory defended by Strunk (1967: 60–61), who maintains that the suffix *-nu-* originates in the reanalysis of the infix to a root ending in *-u* such as PIE **k̑leu-* ‘to hear’ → **k̑l-né-u-/k̑l-nu-* ‘to cause to hear’. However, since this putative proto-type is rare (see also Villanueva Svensson 2025), Strunk had to resort to *u*-extended roots in order to provide further support to this theory (the role of *u*-extensions for this topic is addressed in detail in the last part of this paper). The view presented here also differs from the more recent theory by Steer (2015), who argues that the *nu*-suffix is primary and the *n*-infix secondary. Although his approach is morphologically acceptable, it goes against the evidence of the daughter languages, in which the *n*-infix is older than its various reanalyses as a suffix.

³Thus Steer 2015.

⁴Thus also Oettinger 1979: 192, Shatskov 2020: 281–282.

tarnuwa-ⁱ ‘to release’ vis-à-vis Hitt. *tarna*-/*tarn*-^{hi} ‘to release’ is another example of such a replacement (cf. Melchert 2022: 109–110), comparable to the relationship between Ved. *stṛṇāti* ‘spreads’ and Gk. *στόρνυμι* ‘id.’ or Ved. *yunákti* ‘yokes’ and Gk. *ζεύγνυμι* ‘id.’, where the original *n*-infixing verb has been replaced in Ancient Greek by a *nu*-suffixed verb (see Rasmussen 1999: 308–309 and Villanueva Svensson 2025 for a summary of these phenomena). The radical *n*-infixation in the Anatolian branch is indeed an archaism that can still be observed in Hittite, but almost vanished in the other 2nd millennium BC languages like Luwian and Palaic. In these other two languages, it appears to have been replaced by suffixation after having been reanalyzed with a root-final laryngeal before combining with another morpheme; the result being *-i-na*-ⁱ (consisting of verbal **-ie/o-* or nominal **-i-*, followed by either **-néh₂-/-nh₂-’* or **-néh₃-/-nh₃-’*).⁵ On the other hand, the *nu*-suffix experienced an increasing productivity in Hittite and Luwian, as will be shown in more detail in the next section.

In order to prove that the *n*-infixation among *u*-adjectives is inherited, Koch (1973) cites parallels in Old Indic and Ancient Greek. The Old Indic example for the proto-type that has been cited in the literature for almost a century and was convincingly defended by Oettinger (1979: 164–165) and Koch (1980) is the following one:⁶ Hitt. *tēpu*-/*-aw-* ‘small’ → *teṣnu*-^{mi} ‘to make small, belittle’ matching OInd. **dabh-u-* (replaced by OInd. *dabh-rá-* ‘small’) → OInd. *dabh-n-ó-ti* ‘to harm, deceive’, both leading to the PIE reconstruction **d^h(é)b^h-u/-éu-* ‘small’ → **d^h(e)b^h-n-éu/-u-* ‘to make small’.

Whereas the denominal process has been given up in Old Indic, Koch (1978) and Tucker (1981: 22–28) both argued independently that the process is continued in Ancient Greek factitives in *-ύνω*, e.g., *βαθύς* ‘deep’ → *βαθύνω* ‘to deepen’,

⁵Sasseville 2021a: 540f., 548.

⁶Götze & Pedersen 1934: 69, Kammenhuber 1969: 231, EWAia I 694–696; followed by Oettinger 1979: 164–165, Tucker 1981: 27, Praust 2004: 385–386, Steer 2015: 208–210, Lundquist & Yates 2018: 2162, Oettinger 2022: 157 (adding PIE **méj-u-* ‘few’ → **mi-né-u-ti* ‘to make few’). The first objection raised by Shatskov (2020: 277) that Hitt. *teṣnu*-^{mi} cannot be old because it is not attested in Old Hittite is invalid. The “Old Hittite” corpus is very small and highly fragmentary, which makes Shatskov’s claim *ex silentio*. Shatskov also argues against the word equation due to the, in his words, “quite different” meanings, although Koch (1980: 225–232) has explained in detail the semantic development of ‘to belittle’ → ‘to deceive’ and provided sufficient contextual parallels between the Hittite and the Indo-Iranian cognate (e.g., the gods as patient). It is also interesting to notice that the Hittite factitive *teṣnu*- (just like OInd. *dabhnóti*) shows the restored full grade of the *u*-adjective. Given that the expected zero-grade of the root would have yielded an unstable consonant cluster, i.e., Hitt. [tbnu-], it may have facilitated a restoration of the full grade in analogy to the *u*-adjective in the prehistory of each language. Only Old Avestan *dabənaotā* preserves the original zero-grade with the complex consonant cluster (Oettinger 1979: 164).

θρασύς ‘bold, courageous’ → θαρσύνω ‘to embolden’, εὐρύς ‘wide’ → εὐρύνω ‘to widen’ and *αἰσχύς (replaced by αἰσχ-ρό-ς ‘ugly, shameful’) → αἰσχύνω ‘to make ugly, shame’.⁷ However, Ancient Greek must have innovated by replacing the process of infixation by suffixation followed by the suffix *-je/o- in the present stem of the denominative verbs.⁸ For a new explanation of the development of this derivational process in Greek and its relationship to the denominative suffix -αίνω, see Villanueva Svensson 2024.

2 Deadjectival Verbs in Anatolian

2.1 Formations in -nu-

As is well known, the Hittite language has a significant number of *nu*-formations built from adjectives in -u-, e.g., *parku-* ‘high’ (< **b^herġ^h-u-*) → *park(a)nu-* ‘to make high’, cf. the root verb *parktaru* (3sg.imp.) ‘must become high’; *āššu-* ‘good’ (< **h₁e(h₁)s-u-*) → *aš(ša)nu^{-mi}* ‘to take care of’, cf. the stative, mediopassive root verb *āšš-* ‘to be dear to’; *ḫatku-* ‘narrow’ (< **h₂ed^hġ^h-u-*) → *ḫatg(a)nu^{-mi}* ‘to make narrow’, cf. the root verb *ḫatk^{-hi}* ‘to close’. However, despite what scholars may claim, it is not always clear whether the *nu*-formation is derived from the *u*-adjective or from the root verb.⁹ Although some formations may be more transparent than others, it is always possible to argue for the one or for the other direction, which is relevant for the topic of reanalysis. Considering solely the deadjectival type from a synchronic point of view, one may speak of an *n*-infix or of a *nu*-suffix with deletion of the *u*-suffix. The latter process is precisely the one which applies to the adjectival *i*-stems, e.g., Hitt. *šalli-/ay-* ‘big, great’ (< **solh_{2/3-i-}*) → *šall(a)nu^{-mi}* ‘to make big’, *ḫarki-/ay-* ‘white’ (< **h₂erġ^h-i-*) → *ḫark(a)nu^{-mi}* ‘to make white’, *parkui-/ay-* ‘pure’ (< **perk^u-i-*)¹⁰ → *parkunu^{-mi}* ‘to make pure’ and *mekki-/ay-* ‘many, much’ (< **meġh_{2-i-}*) → *maknu^{-mi}* (*meknu^{-mi}*) ‘to make numerous’.¹¹

⁷Followed by Villanueva Svensson 2024. Earlier works considered the -n- in the suffix -ύνω to be simply an extension without further explanation (see Palmer 1980: 264).

⁸Chantraine 1947: 276–277, de Lamberterie 1990: 25 (“-ῶνω < *-ῶν-ιω, aor. -ῶνα < *-ῶν-σα”). On the denominal nasal suffix in the northern Indo-European languages, see Villanueva Svensson 2016.

⁹Luraghi (1992: 156) sees *aš(ša)nu^{-mi}* as deverbal instead. The etymological rejection of *aš(ša)nu^{-mi}* made by Kloekhorst (2008: 216–218) is rebutted by Sasseville (2021a: 473, fn. 10).

¹⁰For the root, see eDiAna ID 565.

¹¹For a list of examples of the denominal type, see Luraghi 1992: 155–156 and Vanséveren 2014: 70–72.

It is appropriate to recall that there are no clear examples of Hittite *a*-stems, which reflect the merger of **-o-* (unless it was syncopated, in which case the stem became consonantal) and **-eh₂-*, that serve as derivational bases of *nu*-formations. There are a few examples that have been adduced in the literature and the first one is *marš(a)nu^{-mi}* ‘to desecrate’ next to *marša-* ‘false(hood)’.¹² For this reason, Koch (1973: 29) finds its interpretation as a derivative from the *a*-stem *marša-* suspect and therefore does not exclude the possibility that it may have been derived from an unattested root verb or even from a prehistoric *u*-stem. Likewise, Oettinger (1979: 239, fn. 1) and Luraghi (1992: 157) distance themselves from the nominal base *marša-* and presuppose instead a deverbal formation from *maršiye/a^{-mi}* ‘to be false’ with deletion of the verbal suffix *-iye/a^{-mi}* (< **-je/o-*).¹³

A third possibility that has been overlooked is a deadjectival derivation *maršai(ya)-* from an *i*-stem *marši-/ay-* ‘pertaining to desecration, false’, if it truly is a Hittite lexeme and not a Luwianism.¹⁴

Another possible example for an *a*-stem as derivational base is presented in HW ̃ 87–88, where it is argued that *halluwanu^{-mi}* of uncertain meaning is not derived from *halluwā(i)^{-mi}* ‘to fight’, i.e., ‘to cause to fight’, but rather from *halluwa-* ‘deep’, i.e., ‘to deepen’ against the previous literature. The argument rests on the understanding of two contexts, which cannot be reviewed within the frame of this article. Moreover, it is worth noting that the adjective *halluwa-* has a *u*-stem allomorph *halluwaw-* visible in the acc. pl. *halluwawuš* and *halluwāmuš* (see HW ̃ 84–85), which, if this lexeme is indeed the derivational base, may have paved the way for the *nu*-factive, i.e., *halluwaw-* → *halluwa-n-w-*.¹⁵ A similar problematic case concerns Hitt. *hatuganu^{-mi}* ‘to frighten’, which can be derived from the adjective *hatuga-* (or *hatugi-*) ‘frightful’ (Luraghi 1992: 155) or from the root verb *hatuk^{-mi}* ‘to spread fear (intr.)’ (Koch 1973: 19f.; HW ̃ 529), both semantically adequate, i.e., ‘to cause to spread fear’ or ‘to make frightful’.¹⁶ Moreover,

¹²This example is left out in Vanséveren 2014: 70–72.

¹³Note that there is now even less evidence for a synchronic Hittite class of stative verbs in *-e^{-mi}*, since it has finally been recognized that the cuneiform sign E may have the phonetic value of the glide *-i-*; see Kloekhorst 2014: 134–161 and the subtle warnings in the older literature such as in Neu 1983: 10 and Melchert 1994: 35, 141–142. Oettinger (1992: 225–226) has already pointed out that only *ar-ša-ne-e-ši* and *mar-še-e-er* remain as probative examples. Nonetheless, the unmistakable formation *-iye/a^{-mi}* in the amply attested *aršaniye/a^{-mi}* as well as the fact that the hapax *mar-še-e-er* could still reflect a stem *maršiye/a^{-mi}* despite not being written **mar-ši-e-er* render the existence of this class synchronically highly questionable.

¹⁴Cf. eDiAna ID 2869 with further literature.

¹⁵The hapax *maninkuwanut* (CHDS 3.3, 7') is not a relevant example because its derivational base is not (properly) attested (see CHD L–N 170 **maninkuwa-* ‘near’). Therefore, the stem formation of the base is unknown; cf. Oettinger 1979: 246 (“*maninkui-*”).

¹⁶On the few spellings *hatuganu-* for expected **hatuknu-*, cf. the similar situation of *zalukanu-* attested beside the variant *zaluknu-* (Kloekhorst 2008: 1027–1028).

the adjective *ḫatuga-* shows an *i*-stem allomorph *ḫatugay-* in, e.g., abl. *ḫatugayaz*, nom. pl. *ḫatugayeš* (*ḫa-tu-ga-e-eš*) and acc. pl. *ḫatugauš* (HW H 525–529), which may have been relevant for the derivational process, if the adjective turns out to be the proper base. Be that as it may, it is safe to posit that there is no unambiguous synchronic evidence that would show that the Hittite factitive *nu*-suffix can derive verbs from *a*-stems.¹⁷

In the smaller Luwian corpus, we find the several cases of the denominal type in */nuwa⁻ⁱ/* (cf. Sasseville 2021a: 471–477, 484–485) derived from *u*-adjectives, e.g., (1).

- (1) a. CLuw. *wāšu-*, HLuw. *wa/i-su-* ‘good’ (< **h₁uos-u-*) → HLuw. *u-sa-nú-/usnuwa⁻ⁱ/* ‘to make good, bless’
- b. Luw. **/assu-/* ‘good’ (= Hitt. *āššu-* < **h₁e(h₁)s-u-*) → HLuw. *a-su-nu-/ass(u)nuwa⁻ⁱ/* ‘to make good (?)’ (cf. Lyd. *f-ašvo^{-d}* ‘to make good (?)’)¹⁸
- c. HLuw. **/habanzu-/* ‘loyal’ (< **h₂op-ent-ju-*¹⁹) → HLuw. *ha-pa-za-nu-wa-/habanznuwa⁻ⁱ/* ‘to make loyal’ (cf. CLuw. *ḫapanzuwalatar/n-* ‘loyalty’ and HLuw. *ha-pa-zú+ra/i-wa/i-/habanzulawa(i)-/* ‘devoted, devotion (?)’)

From a synchronic perspective, these formations can be interpreted either as *n*-infixes or as *nu*-suffixes via deletion of the adjectival suffix *-u-*, although the former is expected to be the inherited process. The rest of the deadjectival derivations with the Luwian suffix */nuwa⁻ⁱ/* are built on mutated stems that can go back to either a thematic *o*-stem or an *i*-stem, e.g., CLuw. *ura/i-*, HLuw. *MAGNUS+ra/i-* ‘big, great’ (< **h₁ur-ó-*)²⁰ → CLuw. *uranuwa⁻ⁱ/*, HLuw. *MAGNUS+ra/i-nú-wa/i-/uranuwa⁻ⁱ/* ‘to make great’; CLuw. *ḫalāl(i)-* ‘pure’ (Semitic borrowing) → *ḫalalanuwa⁻ⁱ/* ‘to make pure’; CLuw. **zantal(i)-* ‘low’ (< **kmt-ó-lo-*) → CLuw. *zantalanuwa⁻ⁱ/* ‘to make low, diminish’.²¹ As can be gathered from these examples, there is no process of deletion here. Instead, the suffix *-nuwa⁻ⁱ* is added to the thematic vowel *-a-* (< **-o-*), which is the regular derivational vowel in the Luwian *i*-mutated paradigm. In comparison with Hittite, however, one may posit

¹⁷As pointed out by Sasseville (2021a: 483), since the Common Anatolian thematic suffix **-o-* was syncopated in post-tonic position and the formation transferred to another stem class, e.g., *u*-stems, it is always possible that a formation in **-nu-* was originally created at the time when the thematic suffix was still present. Unfortunately, there is as of yet no positive evidence for this claim.

¹⁸See Sasseville 2021a: 479–480.

¹⁹Cf. eDiAna ID 1580 for various ways of interpreting the morphological chain.

²⁰See eDiAna ID 3247.

²¹The lack of a clear example going back to a PIE *i*-stem is simply due to chance.

that the factitive suffix was able to reach the thematic stems in Luwian only because of their merger with the *i*-stems and that the process of deleting the suffix *-i-*, assuming that it existed in Pre-Luwian, was given up, i.e., **-i-* (with deletion) + **-nu-* → **-o-/-i-* + *-nu-* = **-o-nu-*. On comparative grounds, adjectival *u*-stems inherited the *n*-infixation and, as far as we can tell, preserved it.

The next question concerns the path of the extension from *u*-adjectives as the derivational base to *i*-adjectives and the reason behind the reanalysis of the infixation process as suffixation. The two following scenarios require infixation to not only be genuine but also older: The first explanation is that old *u*-adjectives were gradually replaced by younger *i*-adjectives, which motivated the reinterpretation of the factitive formation from infixation to suffixation via deletion of the suffix, i.e., CVC-*u-* : CVC-*n-u-* reanalyzed as CVC-*i-* (ousting the *u*-stem) : CVC-*nu-* (Koch 1973: 35, Tucker 1981: 27). On the other hand, it has been claimed that the reanalysis of the derivational process simply happened due to the decline of the infixation process altogether, which led to a new suffix **-nu-* creating factitives via deletion of the adjectival suffix **-u-* and then **-i-* (cf. Oettinger 1979: 245–247, 2022: 157).²² The reason why the *i*-stems would have been targeted would simply be due to their similarity to the *u*-stems. In the end, the second scenario has the advantage of better explaining why it is the *i*-stems that followed the path of the *u*-stems without having to recur to the loss of an original *u*-adjective.

2.2 New ways of forming verbs from Anatolian *u*-adjectives

The suffix *-nu-* is not the only one that can create deadjectival verbs. At some point, Hittite started using the factitive suffix *-aḥḥ-ⁱ*, which originates in stems in **-eh₂-*, to derive *u*-adjectives via their oblique stem, e.g., *tēpu-/aw-* ‘small’ → *tepawahḥ-ⁱ* ‘to make small, humble’ (vs. *tepnu-^{mi}*), *idālu-/aw-* ‘bad, evil’ → *idalāwahḥ-ⁱ* ‘to treat badly (tr.), to act badly (intr.)’.²³ There is also one potential example in Palaic, i.e., *wāšu-* ‘good’ → *wašūḥ* (2sg.imp.) ‘to act well, be good’.²⁴ Nonetheless, the number of examples is very low, so that the rivalry between *-aḥḥ-* and *-nu-* regarding adjectival *u*-stems is subtle. Shatskov (2020: 282) claims that the two Hittite examples mentioned above (though re-addressing

²²However, the claim found in, e.g., Oettinger 1979: 246–247 that Hittite has stems in *-ui-* (< **-u-ih₂*) is considered outdated here; see Kloekhorst 2008: 829.

²³There is a widespread view that *-aḥḥ-* was mainly derived from stems in *-ant-* via deletion of the suffix (see Oettinger 1979: 240–241, Vanséveren 2014: 147), which would be possible as a secondary phenomenon, but is unconvincing since all examples can instead be derived from the base in *-a-*, which was often extended with the highly productive denominal suffix *-ant-*.

See Sasseville 2015: 284–285 for this criticism.

²⁴See Sasseville 2021a: 536, fn. 8.

only *tepawahh⁻ⁱ*) were created by non-native speakers who did not grasp Hittite morphology, but this is not persuasive. Although there is a strong interference from Luwian native speakers in New Hittite, this is certainly not the case here because the Luwian factitive suffix was restricted to conversion and, therefore, there was no such model for deriving *u*-stems in their own language.²⁵ The Hittite factitive suffix *-ahh-* simply became highly productive and extended its derivational scope to all kinds of nominal stems.

Koch (1973: 29–30) notes that the extension of *-ahh-* to the adjectival *i*-stems is largely restricted to synchronically non-ablauting *i*-stems, e.g., *nakkī⁻*²⁶ ‘heavy, important’ → *nakkiyahh⁻ⁱ* ‘(intr.) to become a concern to’, *šannapili-* ‘empty’ (but cf. acc. pl. [*ša-an-n*] *a-pi-la-a-uš*; CHD Š 159) → *šannapilahh⁻ⁱ* ‘to empty’, *šanezzi(ya)-* ‘pleasant’ → *šanezziyahh⁻ⁱ* ‘to indulge oneself’, *šarazzi(ya)-* ‘upper’ → *šarazziyahh⁻ⁱ* ‘to make prevail’. Regarding the suffix *-zzi(ya)-*, we are dealing with an originally thematic suffix **-tjo-* that secondarily underwent contraction or syncope of the thematic vowel, thus inflecting sometimes like an *i*-stem. A similar explanation applies to stems in *-ili*, whose suffix originates in **-i-lo-* with syncope of the thematic vowel in post-tonic position followed by a secondary transfer of the stem class.²⁷ These data indicate that the Hittite factitive suffix *-ahh-* was still closely associated with the *a*-stems (**-eh₂-*, **-o-*) and, after syncope of **-o-*, with the consonantal stems, after which it was even extended to the (secondary) non-ablauting *i*-stems. An exception to this pattern is *šuppi-/ay-* ‘pure’ → *šuppiyahh⁻ⁱ* ‘to purify’, which would show that the factitive suffix in the end did reach the genuine ablauting *i*-stems, conflicting once more with the factitive suffix *-nu-*.

Overall, the Hittite factitive suffix *-ahh-* was so productive that it could be derived from any substantival or adjectival nominal stem, whereas the factitive suffix *-nu-*, based on the attested data, was confined to adjectival *u*- and *i*-stems.²⁸ The reason behind this distribution may be the nature of the process

²⁵On Luwian interference in Hittite, see Melchert 2005, Rieken 2006, van den Hout 2006, Yakubovich 2010: 15–73.

²⁶For the exceptional origin of this particular *i*-stem, see Widmer 2006.

²⁷See Rieken 2008.

²⁸It is not entirely clear whether Hittite *-nu-* could derive verbs from substantival bases; cf. Vanséveren 2014: 71, who lists three examples: *ešharnu^{-mi}* ‘to make bloody’, *ēdriyanu^{-mi}* ‘to feed’ and *šaminu^{-mi}* ‘to burn (trans.)’, although a deverbal derivation based on the denominative verb *ēdriye/a^{-mi}* ‘to feed’ is another viable option (thus Hoffner & Melchert 2008: 175, fn. 10; cf. Luraghi 1992: 155) which deserves further investigation. The same ambiguity is found in KAL-tarnuškanzi /hadugadarnuskanzi/ (KBo 12.109, 7’), which may be derived from the abstract noun in *-ātar* (i.e., *hatugātar*) or from an unattested denominative in **-iye/a-* with deletion of the suffix. The derivational base of *šaminu^{-mi}* is problematic and thus debatable (cf. CHD Š 118,

itself: Whereas the former is simply suffixed to the nominal stem in Hittite, the latter uses either infixation or suffixation with deletion of the nominal suffix of the base, which must have been morphologically less stable.

There is a view that verbs in *-uwe/a-mi*, which are mostly deradical to roots ending in *-u-* derived with the suffix *-iye/a-mi*, may also be denominal to *u*-adjectives, e.g. *huišu-/aw-* ‘alive, raw’ → *huišuwe/a-mi* ‘to live, stay alive’. Dismissing this morphological interpretation, as well as that of a stative in *-e-mi* because of the lack of parallels, Jasanoff (2023: 69) analyzes this example as a thematized **-u-* present, comparing the reconstructed PIE **g^uih₃ue/o-* ‘live’.²⁹ As per Jasanoff, the evidence for a denominal verb in *-iye/a-mi* derived from a *u*-adjective, albeit formally possible, is lacking independent support. However, what has not been considered is the formal possibility of a deradical verb in *-iye/a-mi* derived from a *u*-extended root. This topic as well as this example is addressed again in Section 3 of this paper.

Turning to the Luwian language, Sasseville (2021a: 190–199) suggests the existence of a denominal suffix */(a)u-di/* used to derive *u*-stems in order to explain the strange morpheme */(a)u-/* found in some Luwian verbs, although a coherent picture of the data could not be successfully provided. Melchert & Yakubovich (2022) take a different approach and convincingly show that the Luwian endings *-unta* and *-aunta* do not stand for the 3pl. preterit, as previously thought, but for the 1pl. preterit ending. Their findings therefore eliminate the evidence for a special denominal suffix */(a)u-di/* in Luwian. In other words, there is currently no other way to derive adjectival *u*-stems in Luwian except with the suffix */nuwa-i/*. As already mentioned, the factitive suffix **-eh₂-* is not a potential candidate either because it is restricted to conversion in Luwian. On the other hand, the Luwian suffix */nuwa-i/* could derive verbs from *a/i*-stems (< **-o-*, **-i-*) and perhaps even from *a*-stems (< **-eh₂-*), thus being more susceptible to encroach on the factitive suffix **-eh₂-*. In the end, this yields a situation that is the opposite of the Hittite one.³⁰

Luraghi 1992: 158). One last possible example of a denominal case is the hapax *aimpanu-mi* ‘to burden’ (HW A 48), if it is derived from the noun *aimpa-/impa-* c. ‘burden’ instead of from the denominative verb **(a)impā(i)-mi* ‘to burden’ (attested solely as a verbal noun *impawar* and as an imperfective *impaiške/a-mi* according to HW I 56). Despite the occasional ambiguity, it does appear as if a desubstantival derivation for *-nu-* was slowly becoming possible, perhaps through reinterpretation of the derivational base.

²⁹See also Villanueva Svensson 2021: 273–274 with fn. 33.

³⁰Cf. Sasseville 2021a: 471–477. However, due to the size of the Luwian corpus and the lack of data, it is not always clear whether the derivational base was mutated (i.e., **-o-*, **-i-*) or not (**-eh₂-*).

Turning to the Anatolian languages of the 1st millennium BC, Lycian has basically eliminated nominal *u*-stems, transferring them to the paradigm of the *a*-stems (< **eh*₂-), so that it does not offer any evidence whatsoever for the current topic. Lydian, however, seems to have had highly productive “*u*”-stems, so that we might find denominal verbs derived from adjectival *o*-stems (< PIE **-u*-), e.g., *taso*- ‘strong, heavy (*vel sim.*)’ (= Hitt. *daššu*- ‘heavy’, CLuw. *kwanzu*- ‘id.’) → *taso*-^d ‘to impose’.³¹ Taking this example seriously, Lydian appears to derive denominal verbs in *-o*-^d of the lenited *mi*-conjugation from adjectival stems in *-o*- via conversion, although the mechanism in this case is more likely to be secondary, rather than inherited.³²

3 Reanalysis and *u*-extensions

According to the analysis of Strunk (1967: 60–108), there is a difference between *n*-infix verbs to roots ending in *-u* (e.g., **k̑leu*- → **k̑l-né-u*-/*-n-u*-) and *n*-infix verbs to roots ending in a laryngeal, in which the *-u*-, unlike the laryngeal, is not part of the root; compare Ved. *śṛṇāti* ‘to break apart’, *śṛṇá-*, aor. *ásarīt*, *śīrti*- ‘breaking’ with Ved. *vṛṇóti* ‘to enclose’, *vṛtá-*, aor. *ávar*, *vṛti*- ‘fencing’. Although Strunk argues that the origin of “*nu*-presents” is to be found in roots that end in *-u*, apart from **k̑leu*- ‘to hear’ and **k̑leu*- (Gk. κίνυται ‘to be in motion’, ἔσσυτο, OInd. *cyuti*- ‘rapid motion’), he only provides examples of these so-called *u*-extended roots, e.g., *dabhnóti* (!), *vṛṇóti* ‘defends’ (cf. Gk. ἐρυσθαί), *vṛṇóti* ‘to enclose’ (cf. Gk. εἰλύω ‘id.’). This demonstrates at the very least that there is a certain connection between *nu*-verbs and *u*-extensions.

The phenomenon of *u*-extension in the Indo-European languages is an obscure topic, especially vis-à-vis the so-called “*u*-present” suffix.³³ García Ramón (2021)

³¹Sasseville 2021a: 180–190; differently Garnier & Sagot 2020: 174, fn. 20.

³²See Sasseville 2021a: 203–205.

³³Cf. Strunk 1967: 60–61, Klingenschmitt 1982: 231–233 w. fn. 4, Steer 2015: 211–217, Jasanoff 2023; cf. also Watkins 1969: 53 on “*u*-endings”. The “*u*-present” explanation is used for Hittite *tarḫu*-^{mi} and *lāḫū*-^{hi} by Oettinger (1979: 222–223., 424). Contra Jasanoff (2023: 67 with fn. 25), Hitt. *tarḫu*-^{mi} can only be interpreted as a root verb of the *mi*-conjugation and the lack of assimilation to **tarruzzi* is due to the generalization of the weak stem, where the accent was not on the full-grade root, across the paradigm (cf. Hitt. *walḫ*-^{mi} ‘to strike’ << **uélh*₂-/*uḫ*₂-). The two attestations *tar-ḫu-id-du* (KUB 36.75 iv 10’) and *tar-ḫu-e-zi* (KBo 38.126, r.col. 10’), which he uses as evidence for an inherited thematic root verb, are not decisive: Both attestations recall the occasional marking of an extra *-i/-e*- vowel following a *-u*-, in this case combined with a marker of the fortis quality of the ending in the first example breaking the consonant cluster, i.e., [tárhh^w,ttu] and [tárhh^w,tsi] respectively; cf. *tar-aḫ-ḫu-i-iš-ke/a-* [tarhh^w,ské/á-] (KBo 61.221 obv.[?] r.col. 2’), *pa-aḫ-ḫu-i-na-li-az* (VBoT 58 iv 36 to ^{DUG}*paḫḫunal(l)i-* n. ‘fire

has recently attempted to differentiate the two by using a semantic approach. He posits that the morpheme *-u-* should be seen as an extension, whenever there is no semantic distinction between the root verb with *-u-* and the one without. However, if the two happen to differ semantically, he claims that the morpheme acts as an aspectual marker (imperfective/perfective), insofar as it appears only in the present stem. If it is not restricted to the present stem, the morpheme is seen as a marker of *Aktionsart*.³⁴ This hypothesis definitely needs more evidence from a larger set of data, which we cannot pursue in its entirety here. The present chapter will primarily be concerned with the morphological aspect of the problem, taking Anatolian “*u*-verbs” as one category. Nevertheless, attention will also be paid to the semantics of the formations in question.

In the Anatolian languages, there are adjectival *u*-stems attested next to root verbs, to formations in **-nu-* as well as to *u*-extended verbs, all built on the same root, and it is tempting to see a connection. A similar picture has already been described for Hittite and Indo-Iranian. Koch (1980: 235) argues based on Old Indic *dabhnóti* ‘to harm, deceive’ next to *ádbhuta-* ‘wonderful’ that the deadjectival *n*-infix formation could at some point be mistaken by native speakers as an *n*-infix verb. In this way, the morpheme *-u-* was reanalyzed as part of the root, i.e., **deb^hu-* or **db^hey-*, and, in a similar way, the Old Indic 8th class with, e.g., *kru-* ‘to do’ may have originated as, e.g., *kr-* ‘to do’ → *krṇóti/krṇvánti* → reanalysis of the root as *kru-* ‘to do’ → *karoti/kurvanti* (cf. Thumb & Hauschild 1959: 265–268 on the problems of this class and its examples).³⁵ Independent evidence for

container’), *wa-at-ku-it-ta* (KUB 30.67, 6’) and *wa-at-ku-at-ta* (KBo 13.137, 11’ to *watku-mi* ‘to jump’; pace Oettinger 1979: 337, fn. 161) and similarly with *i*-glide insertion between *u* and another vowel [*te*]-*pa-u-i-e-eš-ta* (KBo 58.8+ rev. 23’), *ša-an-ḥu-u-i-en* ‘we swept’ (KBo 38.257, 19’), [*ar-k*]-*u-i-an-zi* (KBo 37.70, 6’; HW A 311) and [*a*]-*r-ku-ia-a[n²-zi]* (KBo 40.328, 10’ to *arku-mi* ‘to chant’ attested beside the denominative *arkuwā(i)-mi* ‘to pray, chant’). On the other hand, if one wishes to take these spellings with *-i/-e-* seriously, a deradical verb in **-je/o-* derived from the *u*-extended root with the loss of the *i*-glide remains a viable alternative. In any case, assuming the existence of an additional verbal stem formation which is not securely attested in the first place based on these two forms is highly dubious.

³⁴Unfortunately, the Hitt. verb †*tarḥ-mi* ‘to overcome’ used by García Ramón (2021: 157–158) is a ghostword (outdated Oettinger 1979: 222–223), since only a *u*-extended root verb is truly attested in Hittite, i.e., *tarḥu-mi*; see already Kloekhorst (2008: 837) who convincingly shows that all spellings of *tar-Vḥ-* should be read *tar-uḥ-* not ***tar-aḥ-*.

³⁵Contra Hoffmann (1976), who proposes a special phonological development to derive the *u*-verb from the *nu*-verb, showing on philological grounds that the former has fully replaced the latter over time. One should also note the ambiguity in another example of the same class, i.e., *tan-* ‘to stretch’ → *tanóti/tunvánti* < **tṇ-néy-ti/tṇ-nū-énti* or < **ten-éy-ti/t(e)n-nū-énti*. Assuming the first formation to be original, the morpheme boundary between root and suffix may have been redefined and the derivational process reinterpreted, i.e., as *tan-ó-ti/tun-v-ánti*; cf. Thumb & Hauschild 1959: 265 with further literature. Differently, Jasanoff (2023: 63) takes the middle “*u*-present” to be original (Ved. *tanute*) and the “*nu*-present” to be a backformation thereof.

the validity of the reanalyzed root **db^heu-* comes from Old Avestan *dābaoman-* ‘deception (*vel sim.*)’ with the full grade anchored before *-u-*. According to Koch, the cause behind this reanalysis was the disappearance of the *u*-adjective, i.e., the ultimate derivational base, from the language or languages concerned. However, it is questionable whether this is truly the case and, therefore, it is the goal of the last section of this paper to test Koch’s scenario against a new set of Anatolian data.

The first root examined here is **h₁ues-* ‘good’: A stative verb can be posited on the basis of the CLuw. (though attested in Hittite context) mediopassive form *waššāri* ‘to be good, be dear’ (< **h₁u(e)s-ór + -i*) and the HLuw. active verb *wa/i-sa-* /was(s)-^{ti}/ ‘id.’.³⁶ A *u*-stem adjective is found in CLuw. *wāšu-* ‘good’, HLuw. *wa/i-su* (adv.) ‘id.’ and Pal. *wāšu-* ‘id.’, all going back to a generalized *o*-grade of an acrostatic paradigm **h₁uós-u-/h₁ués-u-*. The crucial *nu*-formation is attested in HLuw. *u-sa-nú-wa/i-* /usnuwa-ⁱ/ ‘to bless’.³⁷ In the end, an apparent “*u*-extended” verb is attested in HLuw. *wa/i₅-sú-ha* ‘I was good, successful’ and probably also in CLuw. *wašuddu*[(3sg.imp.act.?)].³⁸ Unfortunately, due to the lack of attestations, it is not entirely clear what type of conjugation the latter verb has, although, assuming that the Cuneiform Luwian word is complete, it would definitely belong to the non-lenited *mi*-conjugation (just like Lyc. B *welpu*-^{ti}, not ***welpuwe*-^{ti} or ***welpu*-^{di}). The formation in *-nuwa-ⁱ*, despite being primarily analyzed as a deadjectival formation (see Section 2.1), may have been reanalyzed at any time as a causative formation of the root verb. The next question is what lies behind the stem formation of the “*u*-verb”. Sasseville’s explanation as a denominal verb derived from *wāšu-* ‘good’ falls through (see Section 2.2). A synchronic verbal suffix *-u-* is not possible considering the various conjugations shown by the other data (cf. below). What remains is the explanation as a root extension, which would have been triggered by the reanalysis of the causative formation as an *n*-infix stem based on a root ending in *-u*. Moreover, it is striking that the meaning of the verb is basically the same as that of the two radical verbs, which also points towards a simple *u*-extension, i.e., back-projected **h₁uesu-* or **h₁useu-*, backformed

³⁶For a summary of the research on the PIE “stative”, see recently Grestenberger 2019. In the case of *waššāri*, it is difficult to reconstruct a stative of the type “zero-grade of the root with accent on the suffix” due to the expected effects of Čop’s Law in the root, i.e., **éCV > áCCV*. Nonetheless, it is always possible that the shape of the root would have been analogically introduced here after another co-existing stem formation.

³⁷What appears to be a zero-grade might just be a contraction of /wa-/ > /u-/, which sporadically occurs in Iron Age Luwian. Nonetheless, attestations from the earlier stages of the language are required to verify this.

³⁸The Cuneiform Luwian piece of evidence is found in a highly fragmentary context which calls for caution; see Sasseville 2021b: 179.

from the *nu*-formation. If this reasoning is correct, this example alone would contradict Koch's "requirement" for the reanalysis, because both the adjectival base and the root verb without *-u-* are still present in the Luwian language.

The second root adduced here is **h₃er-* 'to arise': This root has a stative verb attested in Hitt. *ār-tari* 'he stands' (< **h₃ér-tor* + *-i*). There are also a few (de)radical formations found in CLuw. *āri-ti* and Pal. *ariye/a-ti* 'to raise' (< **h₃ṛ-íé/ó-*) as well as Hitt. *arāi-/i(ya)-hi* and Lyc. *erije-ti* 'to arise' (< **h₃róih₁-/h₃rih₁-* << **h₃ór-/h₃r-*). A *u*-adjective is attested in CLuw. and Pal. *arū-/aw-* 'standing (straight), tall, high (*vel sim.*)' (< **h₃(é)r-u-/éu-*).³⁹ The well-known causative formation in *-nu-* is attested both as Hitt. *arnu-mi* 'to set in motion, carry off' and HLuw. CRUS+RA/I-*nu-wa/i-* /*arnuwa-i* ' (lit.) to let stand, abandon (?)' and is commonly analyzed as a deverbal formation, although a deadjectival formation derived from the *u*-adjective is also formally possible.⁴⁰ Finally, a "*u*-verb" is attested in CLuw. *aruwarūwa-* 'to lift, erect (?)', HLuw. *261.PUGNUS-*ru-/aruwaruwa-* 'to erect (?)', both reduplicated, and in Lyc. *eruwe-ti*, with a highly debated meaning, but which is interestingly intransitive, thus making the Luwian reduplicated formation a causative thereof.⁴¹ From an etymological point of view, the Lycian verb should simply mean 'to stand' (and the CLuw. verb 'to cause to stand'), although its use with the preverb *hri* 'up' in a context of funerary instructions and prohibitions leaves space for a more specialized meaning. Here again the "*u*-verb" of the *hi*-conjugation is morphologically unexplained so far, although the Greek formation ὀπούω 'to rise quickly, rush away' built on the same root (EDG 1107) might correspond to the *u*-verb found in Luwian and Lycian, i.e., **h₃róu-/h₃ru-*, with the ablaut pattern of the *hi*-conjugation.⁴² Taken together, this would suggest that the root **h₃er-* was reinterpreted as **h₃reū-* in both Greek and Luwian with the root vocalism placed between the last consonant and the *-u-*, as in the Avestan example *dābaoman-* mentioned above. Following Koch's reasoning, this extension would have been caused by the reinterpretation of **h₃r-néu-/nu-* or **h₃ṛ-né-u-/n-u-* (← **h₃(é)r-u-/éu-*) as an *n*-infix **h₃ṛ-n-u-* to a verbal root **h₃reū-*.

³⁹The exact semantics of this adjective are not entirely clear, since it may refer to the king as a favorable attribute (KUB 35.165 obv. 23), to crops (KUB 48.69, 1; here the crops are to grow *arawa* in the manner of a man), and to a tree(?) or better to a door (KUB 35.136+ iv 4', 8', 11', 16', 19', 20'; where the CLuw. ^{GIS}*hīluwa*, which is modified by *arūwa*, may contextually match Hitt. ^{GIS}IG 'door' in line 24').

⁴⁰The HLuw. verb must be derived from CRUS+RA/I 'to stand', as it is made obvious by the logogram (cf. eDiAna ID 2317 on other unpersuasive derivational analyses). The exact meaning of the causative/factitive verb is debatable in the two contexts, especially since it is used each time with a different preverb.

⁴¹Cf. eDiAna ID 2231.

⁴²R. Garnier (pers. comm.).

As a third example, there is the root **serk-* with semantics surrounding the concept of lifting and sublimity found in Hitt. *šarkiške/a-mi* ‘to be sublime’ (< **sṛk-šké/ó-*), an imperfective formation derived from an unattested root verb **sark-*. Hittite also has a *u*-adjective *šarku-/aw-* ‘sublime’ (< **s(é)rk-u-/éu-*) and a *nu*-formation thereof, i.e., *šarg(a)nu-mi* ‘to make sublime (?)’ (< **sṛk-né-u-/n-u-*), although it is attested in a fragmentary context.⁴³ Finally, Palaic shows a “*u*-verb” built on the same root, i.e., *šarku-* ‘to lift’, whose conjugation is however obscured by the lack of at least a 3sg. pres. attestation.⁴⁴ As opposed to the last two examples, the “*u*-verb” is transitive, whereas the root verb, relying solely on its imperfective formation, appears to have been a stative verb. It is not impossible that the Palaic verb *šarku-* was originally intransitive and secondarily acquired transitivity, but without further evidence, this claim is unverifiable. Again, it is worth asking how exactly the *u*-verb was created and if the *-u-* could have been reanalyzed as part of the root on account of the *nu*-formation, provided that it existed also in Palaic or in its prehistory. Furthermore, the *nu*-formation is, with respect to its derivational base, ambiguous, just like in other examples. Both the *u*-adjective (via infixation) and the unattested root verb (via suffixation) are morphologically fine as derivational bases, which again illustrates the constant possibility of morphological reanalysis.

More examples can be adduced, even though not all types of formations are attested, which can also be due to chance. The sole example to be discussed here starts with Hitt. *nāḫ-/naḫḫ-hi* ‘to fear’ and HLuw. **/nah(h)a-i/* ‘id.’ found as the base of the imperfective *na-ha-sa-* (< **noh₂-(h₁)s-*) going back to the root verb of the *hi*-conjugation **nóh₂-n₃h₂-’*. Furthermore, it is attested next to the mediopassive Hitt. *naḫḫantat* ‘they were an object of fear (?)’.⁴⁵ A *u*-stem occurs as the base of CLuw. *naḫḫuwašša/i-* ‘frightening’, which, if it were adjectival, must have been at least substantivized before the genitival adjective in **-oši-* was derived (cf. CLuw. *wāšūwašša/i-* ‘of the goods’). Further evidence for a nominal *u*-stem comes from HLuw. *na-hu-ti-* ‘fright’, which belongs to the nominal type in **-u-ti-* as in Hitt. *-uzzi-*. Unfortunately, a formation in **-nu-* is so far missing from the data, but a CLuw. “*u*-verb” *naḫḫūwa-i* ‘to be an object of concern’ going back to

⁴³Thus also Luraghi 1992: 156.

⁴⁴See eDiAna ID 2736.

⁴⁵Neu (1968: 120) translates it instead as ‘they stood in awe’, thus as showing no semantic distinction to the active root verb. However, since the context is about the capture of Kaskean men, a translation ‘they were an object of fear’ for the *naḫḫantat* is even more probable. However, this cannot be proven beyond doubt, since the context is fragmentary and the restoration of the preceding word [*nu-u*]*n²-[n]a²-aš¹* ‘(to) us’ suggested by Neu does not correspond to the readings offered by the autography KBo 50.16.

a *hi*-conjugated $*n_3h_2\acute{o}u-/n_3h_2u-$ (most likely with a restored full grade between *n* and *h*₂) is attested.⁴⁶ This “*u*-verb”, just like the others, cannot be satisfactorily explained as a denominative via a suffix *-u-* or even via conversion from an adjectival *u*-stem, because the *-u-* does not act like a synchronic verbal suffix, which would have to be restricted to a specific conjugation and the same applies *a priori* for the process of conversion. The data discussed above, however, demonstrate that these “*u*-verbs” can belong either to the *mi*- or to the *hi*-conjugation. In his discussion of PIE *u*-presents, Jasanoff (2003: 141–143) considered the Anatolian evidence to be inconclusive in this respect, while still pleading for the *hi*-conjugation as the original one because of the frequent thematization of these verbs in Indo-European and of Hitt. *lāhū^{hi}* ‘to pour’, despite Hitt. *tarhu^{mi}*.⁴⁷ In a recent article, Jasanoff (2023: 67, fn. 24) withdraws this view and argues instead that *u*-presents were originally proto-middles of the $*e$: zero-ablauting type, which could secondarily inflect as actives with subsequent thematization. From his perspective, the actual conjugation that is attested in a Hittite *u*-verb need not reflect the original situation of the PIE “*u*-present”, which is seen here as a flawed approach. Adding more examples to the list here, e.g., CLuw. *talku^{ti?}* ‘to flatten (?)’ (cf. Hitt. *ištalk^{mi}* ‘to flatten’), CLuw. *malhu^{ti?}* ‘to crush’ (cf. Hitt. *malla-/mall^{hi}* ‘to grind’) and Pal. *āp(p)ūⁱ* ‘?’ (eDiAna ID 1566), simply confirms the reality of the mixed situation, which should be taken seriously.⁴⁸

The synchronic use of both the *mi*- and the *hi*-conjugation is comparable not only to Anatolian root verbs in general, but also to the *n*-infix verbs in Hittite, which may either take the *mi*-conjugation, e.g., *harni(n)k^{mi}* ‘to make disappear’, *šarni(n)k^{mi}* ‘to compensate’, *ištarni(n)k^{mi}* ‘to afflict’, or the *hi*-conjugation, e.g.,

⁴⁶Unfortunately, it cannot be synchronically proven that the ablaut between $*-h_2-$ and $*-u-$ was marked by means of the recurring plene spellings with the sign <u>, i.e., *na-aḥ-ḥu-u-wa* ... (eDiAna ID 3109), since the use of this sign following a laryngeal is a strong tendency in Hittite and Luwian (see Kimball 1999: 67 and Kloekhorst 2008: 51–52). Nonetheless, it is worth noticing the same *plene* spelling in *a-ru-u-ru-u-wa-am-mi-iš* (eDiAna ID 2232), but cf. also the adjective *a-ru-u-* ‘high’.

⁴⁷One should take note that a late *nu*-formation built on the verb matching Hitt. *tarhu^{mi}* has now surfaced in HLuw. TONITRUS-*wa/i-nú-wa/i-tu* /tarhuwanuwa-/ (eDiAna ID 2855). Here, however, the “*u*-extension” is preserved in the base of the *nu*-formation, showing that the *-u-* is understood to be part of the root in Iron Age Luwian.

⁴⁸For a deradical formation in $*-je/o-$ derived from both the *u*-extended and the bare root, cf. Hitt. *šarḫiye/a^{mi}* ‘to eviscerate, maul’ and Lyc. B *zrqqi^{ti}* ‘to maul’ going back to $*s_1rh_3-je/ó-$ and $*s_1rh_3u-je/ó-$ respectively (cf. also CLuw. *šarḫuntalla/i-*, *šarḫuwant(i)-* vs. *šarḫadu*, *šarḫama/i-*, respectively; eDiAna ID 3046). It is also noteworthy that *huišuwe/a^{mi}* ‘to live, stay alive’ (next to *huišu-/aw-* ‘alive, raw’, *huiš^{mi}* ‘to live’ and *huišnu^{mi}* ‘to keep alive, let live’), which was addressed earlier in this paper, may also reflect a formation in $*-je/o-$ derived from a *u*-extended root $*h_2uesu-$ in the same manner as it happened to $*h_1uesu-$.

šunna-/šunn-^{hi} ‘to fill’, *tarna-/tarn-^{hi}* ‘to let (go)’.⁴⁹ Furthermore, as shown in Sasseville 2021a: 481–487, the two conjugational types can be reconstructed for Proto-Anatolian formations in **-nu-* based on the Anatolian comparative evidence.⁵⁰ Taken at face value, this situation brings further evidence to the working hypothesis that the *u*-extensions and the *n*-infixations are ultimately related through a process of reanalysis involving root verbs, *u*-adjectives and *nu*-formations. Moreover, there would be no need to attempt to find out which conjugation is the original one among “*u*-verbs”, since its determination would be parallel to the one among root verbs, i.e., both conjugational types or, in other words, both types of formations can co-exist in the same root.⁵¹

As it could be deduced from the Anatolian data, the requirement that the *u*-adjective must first disappear in the language for the root to be extended with *-u-* in a given example cannot be maintained. At least the first two examples discussed above clearly demonstrate that Luwian can have the reanalysis of the root with *-u-* while preserving the *u*-adjective. This, however, does not falsify Koch’s hypothesis on the origin of *u*-extensions, but rather indicates that the reason behind it must be sought elsewhere. Looking again at the Anatolian examples shown above, the semantics of the “*u*-verbs” seem to match the ones of the stative (or mediopassive) verbs, i.e., 1) CLuw. *waššāri* ‘to be good, be dear’ = HLuw. *wa/i₅-sú-ha* ‘I was good, successful’, 2) Hitt. *ār-tari* ‘he stands’ = Lyc. *eruwe-^{ti}* ‘(intr.)’ with its causative reduplicated stem CLuw. *aruwarūwa-* ‘to lift, erect (?)’, HLuw. *261.PUGNUS-*ru-* /*aruwaruwa-* ‘to erect (?)’ and possibly also 3) Hitt. *naḥḥantat* ‘they were an object of fear (?)’ = *naḥḥūwa-ⁱ* ‘to be an object of fear/concern’. This observation implies that there is a strong affinity in these examples between the “*u*-verb” and the stative verb, which in turn has a close affinity to the *u*-adjective.⁵² In the end, the missing link would be the *n*-infix/*nu*-formation.

⁴⁹For different opinions on whether this situation is adoptable for PIE or whether it is secondary, see recently Melchert 2022: 107–108 with further literature and Villanueva Svensson 2025.

⁵⁰On the problem regarding the conjugation of the suffix **-s̑e/o-* in Anatolian, see Oettinger 2023: 176–177.

⁵¹Jasanoff (2023: 67, fn. 24) sees the mixed situation attested in Hittite as secondary, analyzing, e.g., Hitt. *lāḥū-^{hi}* ‘to pour’ as an originally *h₂e*-conjugated aorist instead of a *u*-present. This approach, however, cannot be synchronically verified. Another idea put forth is that the *hi*-conjugation of **-nu*-verbs comes from the denominal type (just like for the conversion process for stems in **-eh₂-*), whereas the deverbal type was conjugated according to the *mi*-conjugation (Sasseville 2021a: 485–486; Villanueva Svensson 2024, 2025). It is worth asking whether one could extend this pattern to root verbs derived from root nouns by conversion.

⁵²On this type of relationship between *u*-adjectives and stative verbs in state-oriented verbal roots in Indo-Iranian, see Rau 2013.

4 Conclusion

In conclusion, this paper has provided support for the relationship between *u*-adjectives, *nu*-factitives/causatives and *u*-extensions by demonstrating that the phenomenon of reanalysis is responsible for the creation or extension of certain derivational processes. A difficulty encountered in our data concerns the direction of the reanalysis as well as the motivation behind it. To avoid circularity, attention was paid to the rules of word formation to limit the derivational scope. In Anatolian, it could be observed that there is a weak barrier between *nu*-factitives and *nu*-causatives. A given example can be interpreted either as an *n*-infix derived from a *u*-adjective (i.e., *nu*-factitive) or as a *nu*-suffixed formation from a root verb (*nu*-causative) and an objective choice is not always possible. Furthermore, it has become clear that the firmly established denominal *n*-infix in the Anatolian languages began to be reanalyzed as a *nu*-suffix and could thus more easily expand its derivational scope.

Building on the PIE “prototype” $*d^h(\acute{e})b^h-u/-\acute{e}u-$ ‘small’ $\rightarrow *d^h(e)b^h-n-\acute{e}u/-u-$ $\rightarrow$ (root reanalyzed as) $*db^heu-$, the phenomenon of *u*-extended verbs in Anatolian was explored with a selected set of data, in order to understand the origin of the stem formation. The data presented can be understood as exemplifying the same derivational chain as our proto-type. As a consequence, the “*u*-verbs” HLuw. *wa/i₅-sú-ha*, Lyc. *eruwe^{ti}* (with CLuw. *aruwarūwa-*), Pal. *šarku-* and CLuw. *nahhūwaⁱ* were explained as reanalyzed *u*-extended root verbs based on the reanalysis of the derivational base of the *nu*-/*n*-infix formation. This Anatolian verbal stem class and its complexity is therefore effectively comparable to the Old Indic 8th class.

Abbreviations

CLuw.	Cuneiform Luwian	Luw.	Luwian
Gk.	Greek	Lyc.	Lycian
Hitt.	Hittite	Lyd.	Lydian
HLuw.	Hieroglyphic Luwian	OInd.	Old Indic
intr.	intransitive	Pal.	Palaic
KBo	Keilschrifttexte aus Boghazköi	PIE	Proto-Indo-European
KUB	Keilschrifturkunden aus Boghazköi	Ved.	Vedic

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